Section Report 1 — Sequence Similarity, Alignment, and Phylogeny

BIOL 4810

2026-02-17

Section Report 1

Sequence Similarity, Alignment, and Phylogenetic Inference

Purpose: Independently synthesize what you learned about BLAST, alignment, and phylogenetic inference.

Submission: Complete this template, upload required files, click **Prepare submission preview**, then use **Download as PDF (Print)**.

Part 1 — Dataset selection

Gene name:

ribulose-1,5-bisphosphate carboxylase/oxygenase large subunit

Four organisms (one per line):

Vaucheria
Sargassum
Chlamydomonas
Fucus

How is this set different from class examples?

Although they are all algae like the previous sample, the gene I'm focusing on is different, and the type of algae I used has different characteristics. There are some coenocytic algae, green algae, brown algae, and a mix of

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Part 2 — BLAST analysis

BLAST program used:

The NIH Nucleotide BLAST program

Two species selected:

Vaucheria
Sargassum

Interpret E-value, coverage, identity:

E-Value: My results came to an E-value of 0.0. This means that the similarity between my two algae is extremely close to each other and is unlikely to have occurred by chance.
Coverage: My coverage came out to 98%. This means that 98% of my sequence matches the sequences found in the data base.

One conclusion you can and cannot make:

One conclusion we can make is that the Vaucheria and Sargassum are very closely related due to the identity being 81.87% and the E-value being a 0. One conclusion we can not make is that how far ago these algae deviated from their common ancestor.

Upload BLAST screenshot

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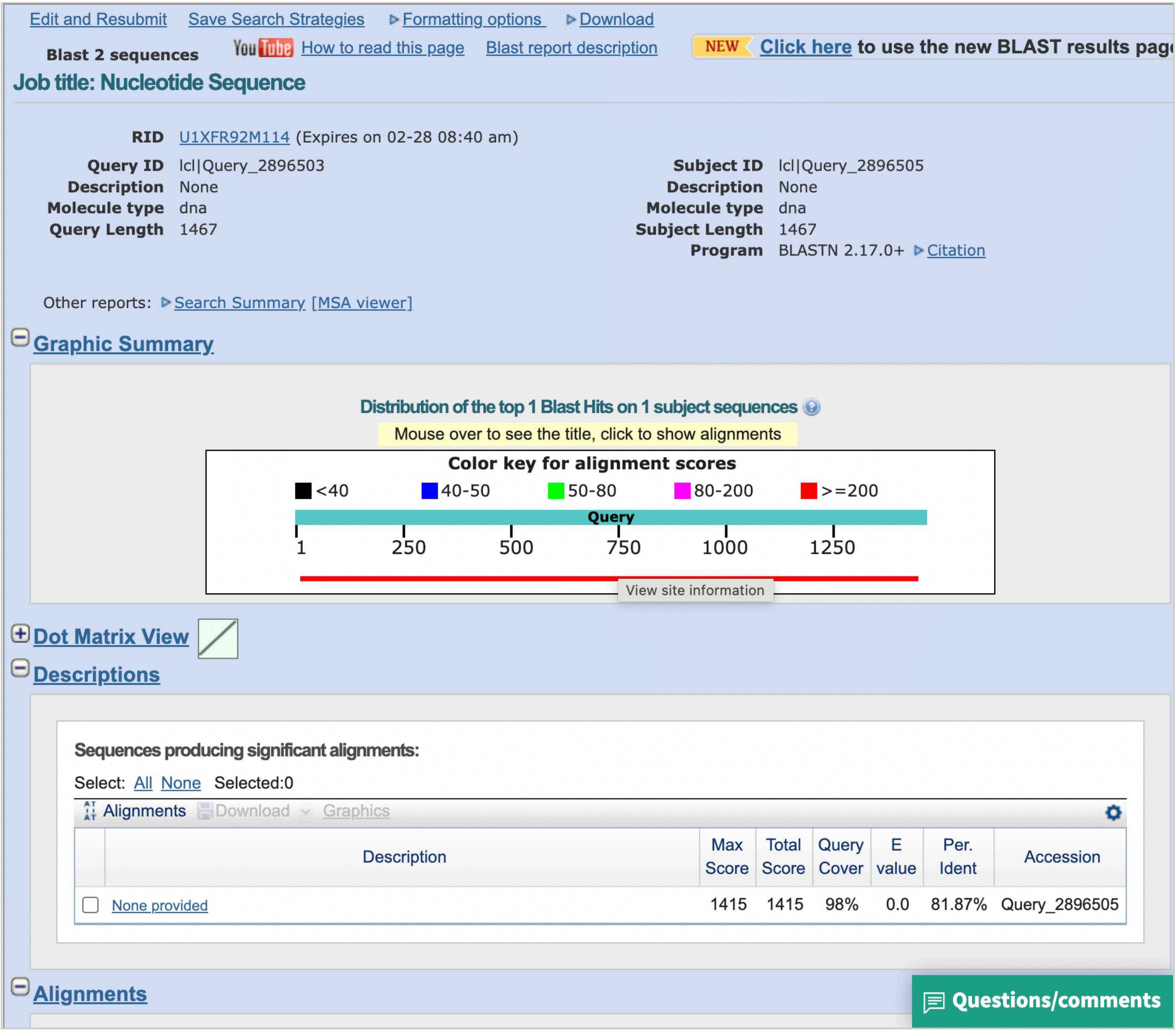
Edit and Resubmit Save Search Strategies Formatting options Download

Blast 2 sequences How to read this page Blast report description NEW Click here to use the new BLAST results pag

Job title: Nucleotide Sequence

RID	U1XFR92M114 (Expires on 02-28 08:40 am)		
Query ID	lcl|Query_2896503	Subject ID	lcl|Query_2896505
Description	None	Description	None
Molecule type	dna	Molecule type	dna
Query Length	1467	Subject Length	1467
		Program	BLASTN 2.17.0+ Citation

Other reports: Search Summary [MSA viewer]

Graphic Summary

Distribution of the top 1 Blast Hits on 1 subject sequences

Mouse over to see the title, click to show alignments

Dot Matrix View

Descriptions

Sequences producing significant alignments:

Select: All None Selected:0

Alignments Download Graphics

	Description	Max Score	Total Score	Query Cover	E value	Per. Ident	Accession
☐	None provided	1415	1415	98%	0.0	81.87%	Query_2896505

Alignments

Questions/comments

Part 3 — Multiple sequence alignment

Aligner used:

T-Coffee

Justification:

I chose T-Coffee for a few reasons. When we did the previous assignment, I had no issues with it and found it very easy/straight forward to work with. It didn't fight me on the documents I submitted, was pretty quick, and it

DNA alignment observations:

The first thing I notice about the DNA alignment is that the alignment result shows a lot of red coloring, meaning good alignments. There is some variable alignments in there too but, only a handful of bad.

Protein alignment observations (BLOSUM62):

I notice that, like the DNA alignment, the protein alignment shows a lot of red/good alignments, with some variable alignments and only a handful of bad alignments. Also, the protein and DNA alignments are very similar to each other.

Upload NESPript DNA PDF

Export/download the PDF from NESPript, then upload it here.

Choose File 0-0-1772158467-esp.pdf

Upload ESPript Protein PDF

Export/download the PDF from ESPript, then upload it here.

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Part 4 — Phylogenetic analysis

Create a Maximum Likelihood tree using your protein alignment.

Conceptual explanation of Maximum Likelihood and how it differs from Neighbor-Joining:

Maximum likelihood uses the most probable events that occurred while Neighbor-Joining looks at the similarities between two groups and joins them that way.

What is bootstrapping?

Bootstrapping rebuilds the tree by resampling the columns. It counts how often clades appear in order to assume the relationship levels. Bootstrapping isn't a 100% accurate method, it's simply an assumption based on previous patterns

Interpret bootstrap values:

My bootstrap value was 100. This means that the program I used was very confident in the tree it gave me and that we can assume that the tree is extremely reliable.

Part 5 — Rooting

After midpoint rooting, describe any changes that have occurred in the tree:

When I midpoint rooted the tree, I didn't see any major changes. The only thing I noticed was that the little tail thingies changed their positions.

What assumption does midpoint rooting make and how is this different from using an outgroup?

Outgroup rooting uses a taxa that is known to not be a member of the main group in order to place a root between the outgroup and ingroup. Midgroup rooting is when a root is placed at the midpoint of the longest path.

Upload iTOL UNROOTED screenshot

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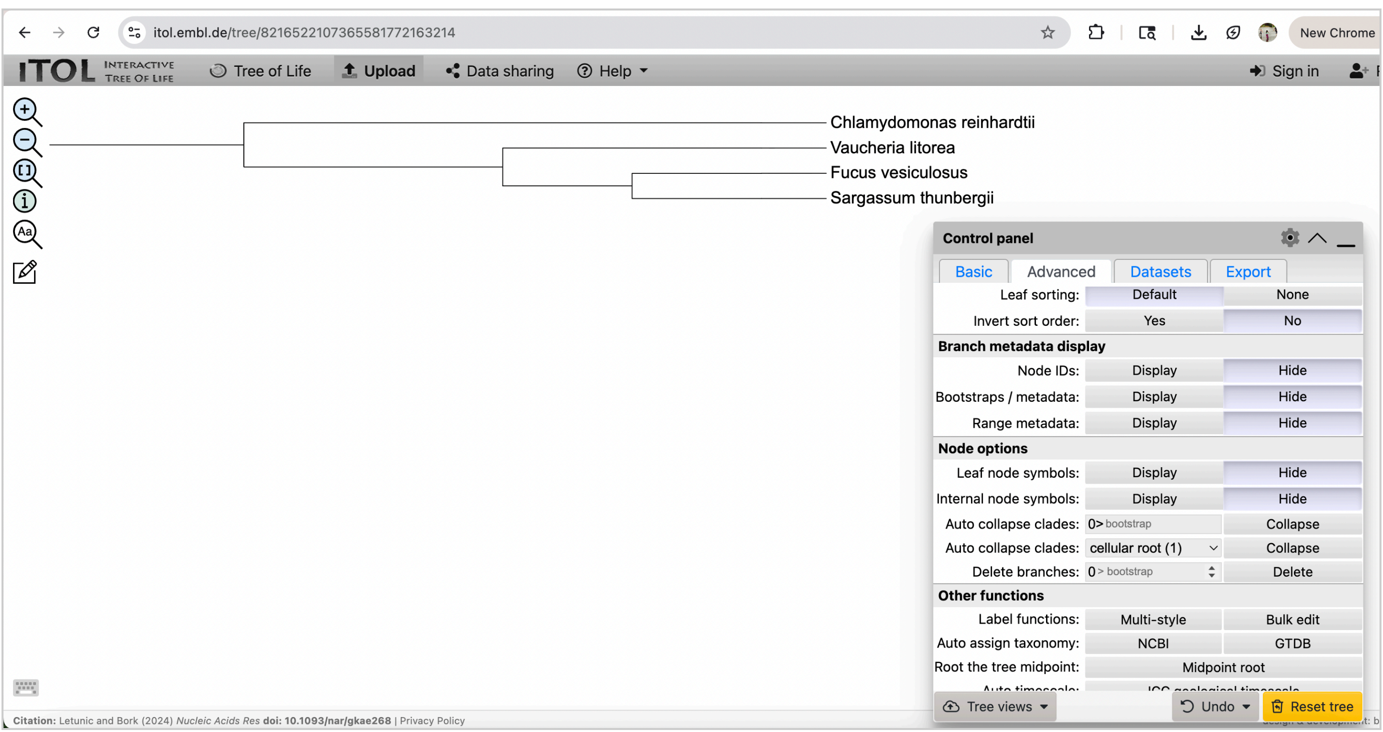

Upload iTOL MIDPOINT-ROOTED screenshot

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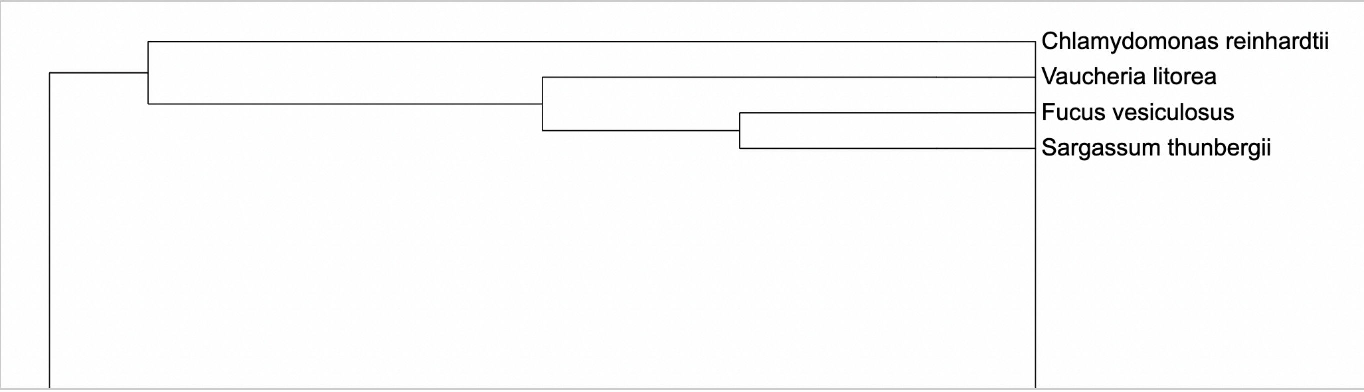

Part 6 — Synthesis

Overall, what can be learned from doing this type of work? Where can uncertainty be introduced and how can that uncertainty be mitigated?

Aligning sequences allows us to compare DNA and protein in order to see how organisms are related and see where their similarities lie and differ. Also, focusing on similarities can give us the percent identity and alignment scores that arise from the inherited traits of a common ancestor. Phylogenies offer us hypothesis about evolutionary relationships based on the shared characteristics and inferred ancestry. It's important to remember that phylogenetic trees are inferences, not facts.

Export